

Room 9 (P1 golden threshold): moderator summary

2026-09-26

Two seats pushed on closing the golden-threshold gap directly.

Tao: genuine Weyl differencing attempted, AMPLITUDE FAILS, and the mechanism is now shown to be the SAME ONE found independently in P2. $\psi_n = [Y^n] \exp(\Phi_q(Y))$ with $\Phi_q(Y) = \sum_{m \geq 1, q \nmid m} u^m Y^m / (m(1 - \zeta_0^{-m}))$ is not slowly varying: the ratio $|\psi_{n+1}/\psi_n|$ jumps $1.6\text{--}4.5\times$ exactly at every $n \equiv 0 \pmod{q}$, riding on an otherwise smooth envelope. Weyl differencing correctly reduces the quadratic phase to a linear one after one step (the phase step works exactly as advertised), but the resulting Abel-summation bound needs the total variation of the shifted amplitude to be small relative to its sum; instead it is dominated by the same periodic $O(1)$ -relative spikes, so no net cancellation survives past the square root. **This is mechanistically the same obstruction Tao's seat found independently for P2's** $w_\rho = c^\rho / (\zeta; \zeta)_{2\rho}$: both amplitudes are built from $1 - \zeta^{(\cdot)}$ factors that are never uniformly slowly varying modulo the relevant period, and in this problem Φ_q is literally log of a q -Pochhammer-type product excluding $q \mid m$, so exponentiating it convolves the near-resonance at $m \equiv 0 \pmod{q}$ into every coefficient. A real, unifying diagnosis across two of this session's hardest problems, not a coincidence.

Erdős: 12 genuine rigorous point-certificates deep in the previously-unreachable window. Three alternative routes (a modular-transform reciprocity, a weaker dense-subset statement, a Jensen/Rouché interpolation-in- ℓ argument) were examined and set aside with stated reasons – none constructed. A fourth, more direct idea worked: using the already-existing rigorous exact-formula tool (`P1e_logderiv.py`, Arb ball arithmetic, no crude bound anywhere), 12 explicit (a, N) points were located and certified throughout $\ell \in (0.082, 0.385)$ – well inside the window the crude method cannot reach – with N up to 701. Every one gives a genuine, machine-verified lower bound on $|Z_N(a/N)|$, e.g. $44/503$ ($\ell = 0.082$, $|Z| \geq 1.528$), $68/701$ ($\ell = 0.183$, $|Z| \geq 0.987$), $48/401$ ($\ell = 0.385$, $|Z| \geq 0.415$). These are narrow

(individual points, not a density statement) but entirely genuine, analogous to P2's individual certified zeros.

Room verdict: ADVANCED. No general theorem for the window and no route to one was found that survives scrutiny – but real content survives: (a) 12 genuine non-vanishing certificates deep in the hard window, narrow but citable; (b) a precise, unifying mechanistic diagnosis connecting this problem's amplitude obstruction to P2's independently-discovered one, suggesting both hard problems in this research program share a common root cause (near-resonant $1 - \zeta^{(\cdot)}$ -type amplitudes defeating standard analytic-number-theory smoothing techniques). The golden threshold $\ell \approx 0.484$ remains the boundary of what the general method proves; the window below it is not empty of results, just not yet covered by a theorem.